

Calculating Limits

Section

1.4

A. Limit Laws

1. Assume that f and g are functions and c is a constant.

$$2. \lim_{x \rightarrow a} (f(x) + g(x)) = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$3. \lim_{x \rightarrow a} (f(x) - g(x)) = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$4. \lim_{x \rightarrow a} (c \cdot f(x)) = c \cdot \left(\lim_{x \rightarrow a} f(x) \right)$$

$$5. \lim_{x \rightarrow a} (f(x) \cdot g(x)) = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$6. \lim_{x \rightarrow a} \left(\frac{f(x)}{g(x)} \right) = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} ; \lim_{x \rightarrow a} g(x) \neq 0$$

$$7. \lim_{x \rightarrow a} (f(x))^n = \left(\lim_{x \rightarrow a} f(x) \right)^n$$

$$8. \lim_{x \rightarrow a} c = c$$

$$9. \lim_{x \rightarrow a} x = a$$

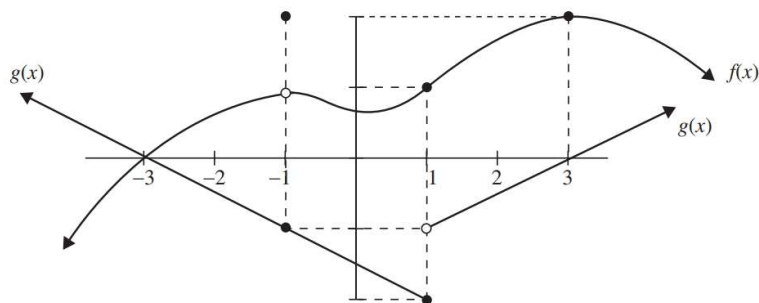
$$10. \lim_{x \rightarrow a} x^n = a^n$$

$$11. \lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$$

$$12. \lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{\lim_{x \rightarrow a} f(x)}$$

Example

1. (video)



a. $\lim_{x \rightarrow 1^+} 3g(x) =$

b. $\lim_{x \rightarrow 1} f(x) =$

c. $\lim_{x \rightarrow 1} (f(x) + g(x)) =$

d. $\lim_{x \rightarrow 3} (f(x) \cdot g(x)) =$

2. (video) Given $\lim_{x \rightarrow a} h(x) = 1$, $\lim_{x \rightarrow a} f(x) = 10$ and $\lim_{x \rightarrow a} g(x) = 0$

a. $\lim_{x \rightarrow a} \frac{h(x)}{f(x)} =$

b. $\lim_{x \rightarrow a} f(x)^{-1} =$

c. $\lim_{x \rightarrow a} \sqrt{f(x)} =$

d. $\lim_{x \rightarrow a} \frac{1}{f(x) - g(x)} =$

e. $\lim_{x \rightarrow a} \frac{g(x)}{h(x)} =$

B. Calculating Limits

Direct Substitution Property: If f is a polynomial or rational function and $a \in \text{Domain}$, then

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Examples

3. $\lim_{x \rightarrow 5} \frac{2x+2}{x-3} =$

4. $\lim_{\theta \rightarrow \pi} \cos \theta =$

Indeterminate Form: If $\lim_{x \rightarrow a} f(x) = f(a) = \frac{0}{0}$ then *factor*, *simplify*, or multiply by the *conjugate*

5. $\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4} =$

6. $\lim_{x \rightarrow 0} \frac{(x+2)^2 - 4}{2x} =$

7. $\lim_{x \rightarrow 0} \frac{\sqrt{x+16} - 4}{x} =$

Undefined Form: If $\lim_{x \rightarrow a} f(x) = f(a) = \frac{\text{Number}}{0}$ then the limit does not exist—**DNE**.

You can support your answer by graphing or using the table method to show.

8. $\lim_{x \rightarrow 5} \frac{x^2 - x - 6}{x - 5} =$

Theorem—We say that a limit exists when the limit from the left equals the limit from the right.

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x)$$

Examples

9. (video) $h(x) = \begin{cases} x^2 + 3x - 1 & \text{if } x \neq 2 \\ x - 1 & \text{if } x = 2 \end{cases}$

Find $\lim_{x \rightarrow 2} h(x)$

10. $h(x) = \begin{cases} x^2 & \text{if } x \neq 2 \\ -1 & \text{if } x = 2 \end{cases}$

Find $\lim_{x \rightarrow 2^-} h(x) =$

$$\lim_{x \rightarrow 2^+} h(x) =$$

$$\lim_{x \rightarrow 2} h(x) =$$

Theorem—Squeeze Theorem: If $f(x) \leq g(x) \leq h(x)$ and $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$ when x is near a , then $\lim_{x \rightarrow a} g(x) = L$

Examples

11. Find $\lim_{x \rightarrow 0} x \cdot \cos\left(\frac{1}{x}\right)$

More Examples

12. $\lim_{x \rightarrow -8^-} \frac{5x + 40}{|x + 8|} =$

$$\lim_{x \rightarrow -8^+} \frac{5x + 40}{|x + 8|} =$$

$$\lim_{x \rightarrow -8} \frac{5x + 40}{|x + 8|} =$$

13. Evaluate $\lim_{x \rightarrow 0} \frac{\tan(5x)}{\sin(6x)} =$